(341)

REPORT OF TRIP, 6-11 April 1955, to:

[redacted]

GENERAL

Between 6 April and 11 April 1955, the writer visited certain cleared contacts at the [redacted] (7 and 8 April); talked to [redacted] at the [redacted] (9 April); talked to [redacted] consultant at [redacted] (10 April); and talked to [redacted] at the [redacted] (11 April). Details follow:

On the afternoon and in the evening of 6 April, the writer and [redacted] Chief, CS Contact Office, [redacted], discussed the [redacted] and the personalities scheduled for conferences the following day. [redacted] and the writer left [redacted] in a company car about 8:00 AM, 7 April and arrived at [redacted] (75 miles southeast) at approximately 9:45 AM. Shortly after arrival, conferences began. At the first conference, in addition to [redacted] and the writer, there were present [redacted], Chief, Civilian Administrator [redacted] and [redacted], Chief of the Medical Research. Following pre-arranged plans, the writer after giving [redacted] security briefings, outlined in general terms the ARTICHOKE work and matters of related interest. This conference lasted approximately two hours. After lunch, the same briefing was repeated with the above-mentioned group plus [redacted], Senior Consultant and former Chief of Anesthesiology and [redacted], Chief of Psychiatry, present. This conference, including general questions, lasted about three hours. Later that afternoon at the [redacted] home of [redacted], [redacted], [redacted] and the writer continued the discussions for several additional hours.

[redacted]

On 8 April, between 9:00 and 11:00 AM, [redacted] and the writer were shown over the physical layout of [redacted] and the new Medical Sciences Building, including detailed visits to the [redacted] library, operating rooms, laboratories, animal cages, engineering set-up, radio-active areas, communications area, files, security systems, etc. At 11:00 AM, another general conference was held with [redacted] and the writer present. Following lunch, the writer had several short individual talks with Mr. [redacted] Dr. [redacted] and Dr. [redacted] At approximately 1:30 PM, [redacted] and the writer left [redacted] and returned to [redacted]

AT [redacted]

During the two-day visit [redacted], the writer was able to introduce for discussion many subjects of interest to ARTICHOKE and other programs of interest to us. These discussions, while mostly general in nature, were at times sufficiently pointed to obtain some specific answers or opinions. Set out below are statements in substance concerning various items discussed:

LSD

[redacted] particularly stated that this was a very important chemical and perhaps with study the most valuable for our use. He, himself, is doing experimental work with it but stated at the present time the control of the chemical for our use has not yet been worked out. He agrees the chemical has a very high potential and much experimentation is needed. He has not used it in psychiatric case-work.

[redacted] (An ergot derivative -- and possible antagonist of LSD)

RAUWOLFIA

[redacted] consider this a very interesting chemical. They do not believe it would be valuable by itself for our use but feel that it would be extremely interesting worked in combination with other chemicals. [redacted] particularly thinks it should be tried with Chlorpromazine and possibly also in combination with LSD.

BULBOCAPNIN

[redacted] stated that there was interesting material in Bejong's book on this chemical and he thought the psychogenic effects might be interesting although he had not worked with it.

NERATRAN

It was suggested that Veratril (the gamma isomer of Neratran), Merrill product, would perhaps be more effective and interesting than Neratran. They suggested study also along these lines.

POLITRONE

[redacted] stated that in his professional opinion, Politrone is highly useful for its analgesic effect. He stated that obviously there was a psychogenic effect in use of the chemical but because of the nature of his work (anesthesiology) he had not worked with it experimentally along these lines. He also indicated he felt that judging by the several hundred cases he had worked with, it might have an amnesic effect. [redacted] both stated that it was important and should receive much consideration. [redacted] stated the chemical should be extremely valuable for the Agency and cited this example: A group of [redacted] roughhousing in

one of the locker rooms of a nearby country club and one doctor twisted his back and neck in such a way that he could not straighten up, every movement being very painful. [redacted] brought his back to the Clinic and gave him 10 cc. of Dolitrone intramuscularly in the buttock and within minutes this doctor was able to straighten up, move about , even using a golf club and felt no pain. [redacted] said he did not check out how long the condition lasted but he knew it lasted at least several hours and he stated it was the first time Dolitrone had been used intramuscularly. [redacted] felt that for people who had to jump from parachutes and in line of duty be subject to pain, broken legs, sprains, etc. and who had to perform an assignment in a given period of time, this might be an important discovery. The writer suggested that this could be worked out using an automatic injector device and [redacted] said in his opinion it definitely could. He stated he would work on the problem and give it thought since he clearly recognized its value to us. He and [redacted] agreed that they would do some work in the future on the psychogenic and amnesic effects of Dolitrone.

MESCALINE

[redacted] advised using Mescaline in combination with other chemicals. He stated it should be tried with Rauwolfia and LSD. He said he had done no experiments along these lines but thought it might be important.

XENON (a gas)

[redacted] knew of this rare gas and knew that it had been studied in the U.S.S.R. [redacted] admitted that it was a powerful anesthetic gas but felt its expense and way of administration might preclude our use of it.

[redacted]

ASPIRIN

Both doctors agreed that the aspirin tablet was a simple way of administration and possibly aspirin combined with other chemicals might produce interesting results. They had done no work along these lines but admitted that pills could be made to resemble aspirin tablets and this was indeed a very common pill.

MARIJUANA

Interesting but did not appear to be good from our point of view. [redacted] doubted its value as a talk producer particularly since control would be difficult. [redacted] suggested experimentation with this combined with some other elements.

BARBITURATES

Doctors [redacted] are, of course, familiar with all types of barbiturates. [redacted] has used Barbiturates in psychiatry to a great extent. Use of Barbiturates, of course, as an anesthetic is standard and [redacted] has made tremendous contributions in the field of Barbiturates, particularly his introduction of sodium pentothal. The writer asked both men if they were familiar with any new chemicals in this field with unusual and strange results. The writer was informed there were no particularly new barbiturates currently being studied and any new information along these lines in the future would be transmitted to the writer. [redacted] agreed with the writer in the use of certain barbiturates coupled with certain stimulants in narco-interviews as being effective. He has had considerable experience along this line.

[redacted] said he would give the matter some consideration and would also discuss the problem with [redacted] (not yet cleared).

TERTIARY AMYL ALCOHOL

The writer asked [redacted] and [redacted] if they were familiar with it and [redacted] said he didn't think it was worth anything and was quite familiar with it. He stated a compound known as "HY2" put out by Parke-Davis was far superior. (The writer may have misunderstood the term "HY2" but since it is contemplated further work with these men this matter will be taken up at that time.)

PIPRADROL

Term mentioned but no comments made. Apparently this compound may be experimental and hence not yet used clinically.

SERATONIN

Both doctors were familiar with Seratonin and [redacted] particularly described it as very interesting but he had not done any work with the material. [redacted] stated that he would examine what he had on Seratonin and that at some future date the matter should be taken up with him again. He stated obviously it was of great interest but he did not feel at the present time it was worthwhile for him to comment upon its applications and results.

DIBENZYLINE

[redacted] stated it was no good, from his personal knowledge.

[redacted]

ACID, LYSERGIC

[redacted] commented that these peculiar compounds often had interesting psychogenic effects. They stated that they had not studied it but doubted it would have much use for us. This was apparently based on the unusual side effects which probably would not be useful and might definitely be dangerous.

CHLORAL

[redacted] stated he thought that Chloral Hydrate was not good and in his use had never been successful. However, he did not preclude work with it but did not like the substance.

DOBUSON

[redacted] both stated this was no good.

ANTIHISTAMIC DRUGS

Both doctors recognized that these drugs would produce psychogenic effects but offered no comment.

FUNGUS TOXINS — MUSHROOMS

One conference had hardly started before Dr. [redacted] asked the writer if he were interested in poisonous mushrooms. The writer informed Dr. [redacted] we were studying them now. He urged a very careful study be made of this and also of the literature as it appeared that toxic mushrooms were used as intoxicants in the Soviet Union. He felt it was something that had not be exploited and suggested study as to the effects produced by these agents.

MORPHINE

Again both doctors were well acquainted with Morphine and suggested that further exploration could be made in using Morphine in combination with such elements as scopolamine and others. (No special suggestions made.)

BARBITE, PENTOTHAL

The doctors were familiar with the name but had not worked with this chemical.

COMBINATIONS

Dr. [redacted] stated that he felt that special attention should be devoted to the use of Serpasil (Rauwolfia) with the relatively new chemical Chlorpromazine. He stated he had not as yet experimented in this connection but felt it might be valuable. He called attention to an editoral in the ANNALS OF INTERNAL MEDICINE, CURRENT ISSUE, an article on psychogenic drugs. He suggested it would be of interest.

Dr. [redacted] stated that at [redacted] there was a pharmacologist or chemist whose name he could not remember who was specializing in studying analogs of LSD, and had developed well over 100 of these analogs. He suggested this man be cleared and interviewed.

Dr. [redacted] stated we should clear or arrange to contact Dr. [redacted], United States Public Health Service, Baltimore, Maryland. He

Stated [redacted] was a brilliant gadget man but possibly hard to control. He stated that [redacted] developed the plastic needle. [redacted] stated he thought something could be done with the insertion of a plastic needle into a person's veins that would be an ideal type of lethal weapon for self-destruction. The writer did not develop this idea since entre to [redacted] is assured in the future and this field, while of interest, is not particularly pertinent.

ADRENOCHROME

The writer mentioned Adrenochrome and related substances as possibly psychogenic agents. [redacted] agreed that they were possibly of interest but stated he was not exactly certain how this could be used.

HYPNOSIS

The writer introduced the subject of Hypnosis and explained in general terms our interest in it and our work with it. [redacted] of course, was familiar with hypno-analysis but is not using the technique apparently at the present time. He has not tried using any of the new chemicals as a trigger for Hypnosis and to his knowledge no one has tried obtaining hypnosis in the semi-coma period following shock-therapy. He agreed very little was known about the actual ultimate ends that were possibly obtainable through hypnosis. He stated that it certainly deserved intense experimentation and thought experiments could be designed to establish how far an individual could be pushed under hypnosis.

DFPP

[redacted] made reference to DFPP by initials. Since the discussions at times were highly technical between various individuals present and since time was limited, the writer could not trace out all these various chemicals mentioned. Later at [redacted] the writer asked [redacted] what DFPP was and its use and [redacted] referred the writer to the 1955 Edition of GOODMAN AND GILMAN, page 458.

COCAINE

[redacted] were quite familiar from a clinical point of view with Cocaine and its derivatives. [redacted] agreed that work could be profitably done in this field.

GLUTAMIC ACID AND RELATED SUBSTANCES

[redacted] stated he was familiar with this and realized that it would have psychogenic effect but did not comment at length on it.

ALCOHOL

[redacted] agree that Alcohol, particularly in whisky, wine, etc. was highly important to any operations of an Agency of our type. They suggested that practically everything mentioned should be worked in combination with Alcohol and that they would themselves try combinations with it and would examine possible chemicals that would work as potentiators for the effects of alcohol.

ACTH, COMMISSION

STOMACH TUBES

[redacted] stated that the Swedes have a way of placing oxygen via a tube into the mother's stomach to assist in childbirth. He stated oxygen is pumped into the mother's stomach through a tube and instead of having a blue baby at birth, apparently a pink baby results. This has been used successfully. He suggested we give some thought to placing gas into a subject's stomach as a possible way to produce talk. He said different gases could be induced into the stomach through a tube and absorbed from the stomach into the body. [redacted] said this could be easily accomplished with an unconscious subject.

ELECTRONIC DEVICES, ELECTRONARCOSIS

All persons that were talked to or sat at t[redacted] stated that [redacted] had a vast array of electronic equipment of all types. They stated they built their own devices if devices were not available on the market and at [redacted] they had numerous devices that were entirely unique. Most of these devices would be too complex, too clumsy or require too much attention to serve a useful purpose in our work. [redacted] stated he was quite familiar with electronarcosis and stated it was clumsy and painful and felt it would not be useful to us.

It should be noted that the group talked to stated that they would be glad to design and build apparatus for us if such were required or that they would themselves attempt to develop apparatus or devices that arose fr[redacted] their thinking along lines mentioned heretofore.

UNUSUAL NEEDLES

The writer stated we were extremely interested in any unusual

results of new chemicals or strange results that arose from use of new chemicals, combinations of chemicals, electronic devices, UHF sound, etc.

All individuals talked to stated they were well-aware these things occurred but at [redacted] they were usually quickly eliminated since they would not be in the best interests of the patients. They said they would have to think this problem over before appropriate answers could be made.

DIETARY DEFICIENCIES

The writer introduced the matter of dietary deficiences as a possible weapon. The writer pointed out a standard Soviet or Communist technique in brainwashing or in prisoner-handling appears to be a barely life-sustaining food ration. The writer asked the individuals he talked to to consider this problem and whether they thought it might be possible to manufacture food which to all outward appearances would look and taste normal but would, in fact, be specially prepared (vital elements removed) and would create confusion, anxiety, fear, etc. in a prisoner or perhaps lower mental resistance, increase suggestibility or affect the memory. All doctors talked to felt this was too complex and in their opinion it would be much easier to add something to the food for these purposes than create new foods. [redacted] suggested that [redacted] be consulted in this matter. He stated that [redacted] two-volume book on diet deficiency would be of interest to us.

It should be noted that [redacted] retired specialist on internal medicine and authority on emotions and suggestions and their

on the digestive system is available at [redacted] for consultation but he is not cleared and was not talked to although it appears that he might be valuable in this connection. [redacted] is also interesting in that he had done a great deal of work on pain and its effects.

POTENTIATING AGENTS, GENERAL

Potentiating chemicals, the writer pointed out are interesting to us. All doctors agreed that this field should be explored. They have nothing to offer of particular interest at the present time and said it would have to be given thought.

WYAMINE SULFATE

[redacted] suggested that consideration be given to our use of Wyamine Sulfate produced by Wyeth Corporation, Philadelphia, Pennsylvania. This chemicals has about the same effect as Benzedrine only it is stronger and more effective. They suggest persons in our work under intense pressure and suffering from strain or hangover might find this preparation useful. They stated, however, it was probably well known to us.

BRAINWASHING

For general purposes, the writer introduced the subject of Brainwashing. All individuals talked to expressed great interest in the matter but felt that their thoughts were not solidified enough to advance any ideas in this connection. They could suggest no techniques or subtle chemical treatments that might have produced this phenomena but st

they were not informed in the matter except through the public press.

[redacted] urged that we clear and contact [redacted] whose specialty was anesthesiology and who at present is on the staff at [redacted]. [redacted] stated that [redacted] was a very brilliant individual who was constantly dreaming up odd or unusual ideas and manufacturing unusual gadgets for use in medical work. He thought he would be an ideal type for us to talk to.

UNWITTING OR UNWILLING SUBJECTS

The writer discussed at length the problem of the unwitting and even the violently unwilling subject and asked for some opinions or suggestions as to how such subjects could be brought into a controlled situation. This problem, according to all interviewed, required considerable thought. Closely related to the above was the problem of what could be done to a person who was asleep without disturbing and without leaving telltale marks on the person or leaving a hangover. The answers were again the same but [redacted] showed the writer a very small sized entirely portable machine for administering gas which had possible use in these cases. This small highly portable device had containers of various gases in little cylinders similar to those used in the sparklet soda bottles. [redacted] suggested that this machine seemed extremely valuable for our study (see literature attached). As far as is known, this machine has not yet been marketed and the writer is of the opinion that [redacted] had the first two machines in his office for study. [redacted] also pointed out that as a first-aid device or an oxygen device when working with

barbiturates this device had great merit particularly under field conditions.

PSYCHOLOGICAL HARRASSMENT

The writer asked various contacts at [redacted] if they had made any studies of psychological harrassment of any type. The answer was negative. The writer asked if any work had been done on such things as unusually painted rooms or slightly out-of-proportion rooms, rubber rooms, vibrations, smells, etc. The answer again was negative. This, obviously, appeared to be based on the fact that [redacted] was for assisting human beings who are ill and not for research into psychological problems.

LOBOTOMIES, ETC.

In the discussion concerning lobomories, it was ascertained that [redacted] neurosurgeons performed lobotomies of various types as a matter of course. They were not particularly impressed by the use of lobotomies for our work based primarily on the fact that it was clumsy, dangerous, required expert surgery with not too predictable results. They were aware of the term chemical lobotomy but apparently felt that this was not suitable. They knew of brain surgery performed by [redacted] sound but since this required opening the skull they thought this also was not feasible. ~~(See Dr. [redacted])~~

MEDICINAL PLANTS

The writer requested if in [redacted] research they had discovered or worked with any "unusual" medicinal plants or had studied the eff

effect of "unusual" plants upon humans. In this connection, it is interesting to note that [redacted] again brought up the subject of mushrooms, particularly the toxic variety. [redacted] stated he was studying mushrooms in a general sense and had noticed several articles in the literature describing the use of mushrooms as an intoxicant, particularly among primitive Siberian tribes. He thought this was very important and felt a study should be made. He suggested that perhaps mushrooms which were toxic possibly used at dinner if worked in combination with LSD or mescaline might produce unusual results. Details not developed.

TOTAL ISOLATION

Perhaps the most interesting discussion of the entire trip occurred during one of the conferences. [redacted] without any previous reference, asked the writer if he were familiar with the work of Dr. HEBB in Canada. The writer stated he was familiar with HEBB's work. [redacted] then asked did the writer note the significance of Dr. HEBB's work from an intelligence point of view. The writer replied that the matter was known and we were examining it and studying it as best we could. The writer asked [redacted] if he had done any work along these lines and he replied he had not but had read all available literature along these lines and wanted to do some experimentation if he could make some arrangements. [redacted] stated Dr. HEBB's work was quite well known and stated the effect on animals was also fairly well described in the literature. [redacted] then stated that this field should definitely be exploited. He stated it was of extreme importance and that it might be a very important intelligence weapon. He urged that worked be done in this connection and the isolation techniques be used in combination with chemicals such as LSD, mes-

calling, Rockville, etc. The writer discussed the isolation problem in very general terms with Dr. [redacted] and particularly asked them to give consideration to ways and means to which the isolation technique could be brought to bear on willing or more important unwilling or even violently-objecting subjects. [redacted] stated that the problem appeared to him of tremendous importance and he certainly intended to give this matter considerable thought. [redacted] urged complete studies of experimentation and research in this field a number of times during the conversations in the several conferences the writer had with him.

POLYGRAPH

The writer in very general terms discussed the polygraph and some of the uses made of it. Whereas most of the doctors present had heard of the device, none of them had had a chance to see it in actual work although [redacted] had worked with Dr. Leonard [redacted] in connection with the original development of it. In response to exact questions concerning measuring devices which would be of use to the Polygraph Branch, the writer was informed that there were on the staff of [redacted] several individuals whose names will be mentioned later who had done very advanced work in the inventing of and development of sensitive measuring devices for blood pressure, skin radiation, etc. One doctor is supposed to have developed numerous pleythysmographs and related measuring devices which are capable of adaption to the measurement of decreased or increased blood pressure or blood flow at many points in the body. This discussion was not developed because the specialists referred to were not cleared and it was felt that exploration along this line could wait for th

these clearances.

INHALANTS OR NEW ANESTHETIC GASES

Dr. [redacted] stated at the present time there were no new inhalants or gases he felt would be of interest to us. He pointed out that modern surgical techniques of anesthesiologists require complex hospital-type equipment and all modern operating rooms are extensively equipped but that this type of equipment could certainly not be transported into any crude area or used under field conditions.

TWITCHES, JERKS, SHAKING, CONTROLLED PRODUCTION OF

Since the writer and Dr. [redacted] had agreed that the production of any of the above phenomena would be of considerable interest if they could be controlled, the question was put to the doctors interviewed. While they regarded the problem as very interesting, no answers were obtainable. Dr. [redacted] suggested that ~~Dr. [redacted] Professor of Psychiatry at the University of [redacted]~~ should be cleared and interviewed. Dr. [redacted] stated that Dr. [redacted] had for many years studied and done research in unusual types of psychological phenomena and he probably would have a considerable amount of material along these lines.

TOXICOLOGY

The writer ascertained that [redacted] does not have a toxicologist as such but they do have chemists and pharmacologists on their staff who have done toxicological work and their services would be made available at any time a toxicological study was required.

PAIN-PRODUCING AGENTS

The matter of the controlled production of headaches, earaches or various pains was discussed with the cleared contacts at [redacted]. They agreed that certain pains and aches could be produced by chemicals but the controlling of these as to length of time of the attack and time before the onset of the attack would be difficult. The problem of how to produce a specific illness or pain that would develop several hours after the giving of a chemical was discussed but no immediate ideas were proposed in this connection.

"AUTOMATIC ANESTHETIC MACHINES" ([redacted])

This topic was introduced for discussion but since the machine was described as extremely complex, cumbersome and utterly useless under our circumstances the matter was not pressed.

MICRODISCHARGE DEVICES

In view of the failure of the hypospray to be of particular value to us, the Subject of Micro-Discharge devices for injection of powerful new agents was introduced. [redacted] stated he knew of the pencil injector-type of device but stated as far as he knew no one had built any of these devices on a very small scale. He agreed, however, that with our knowledge of small hypodermic needles and injector mechanisms such a device could be built. [redacted] stated this type of device would be effective for lethal purposes. He stated this type of device could be built in small sizes that could definitely penetrate heavy clothing, including leather jackets, etc.

[redacted]

AMNESIA

The writer went into some discussion about the problem of a controlled amnesia and its value. The writer explained that our experiments indicated that certain chemicals, hypnosis and other things produced amnesia but these were not controlled, were not generally total and that under no circumstances would we guarantee amnesia in any type of work. The doctors discussed the matter and agreed that controlled amnesia or total amnesia was not obtainable except possibly through surgery which they ruled out. They also felt that electroshock or chemically-induced shock, while they would generally produce amnesia for the actual shock operation, would not produce amnesia for periods preceding the administration of the shock. They indicated there was some amnesia produced for considerable periods following the shock but this certainly was not controlled and would be spotty.

[redacted] stated he felt if he were given the problem he would try to produce an amnesia through the use of a series of differently-acting psychogenic chemicals or chemicals that act upon different areas of the brain. He stated he thought that by switching the chemicals back and forth both prior to the interrogation, during the interrogation and following the interrogation it might be possible to produce a form of amnesia. He stated that certainly a subject who had undergone various types of reactions to certain psychogenic chemicals and had been interrogated along with these various chemicals would be extremely confused, probably disoriented and would not have sufficient mental capacity to differentiate between fact and fancy.

Neither doctor doubted his motives or information.

HYPOSPRAY

The writer asked the doctors (particularly [redacted]) if they had made use of Hypospray devices. The doctors were familiar with the device but apparently no use was being made of it at [redacted]. Dr. [redacted] stated he did not like it primarily because it produced bruises and wounds and he felt for his work that other means of administration were more suitable and more adaptable. They knew of no new hypospray devices and certainly were not designing or developing any themselves. Dr. [redacted] pointed out that a Hypospray could not safely be used through clothing or leather. They were not familiar with Dr. [redacted] work at the University of [redacted] (sub-lethal shots of cyanide by a "new" hypospray as a prisoner-control device.).

NERVE GASES

The doctors at [redacted] were familiar with the various nerve gases but such elements, which have no real clinical use and are highly dangerous, are not handled at [redacted]. The doctors admitted that these gases or gases which could attack the nerves in sub-lethal doses were interesting but no suggestions were offered.

CONVULSIONS

The writer explained our general theoretical interest in the production of convulsions as a direct weapon or possibly the use of the coma period following the convulsion as a means of obtaining hypnosis.

[redacted]

The writer also discussed flickering lights or interrupted sound as a method for setting off a convulsion in a chemically-conditioned subject. The doctors, while familiar with convulsions and shock therapy were unable to offer specific suggestions although they agreed the ideas were interesting. They suggested the following:

a) Ammonium Chloride

Ammonium Chloride intravenously will produce convulsions.

b) Pyradoxin deficiencies will cause convulsions.

c) Trizol (an analog of Metrazol) is a convulsant. Known as Trizol in Germany in 1938 and 1939.

d) Diberal, a barbiturate of the Eli Lilly Company is a convulsant.

d) Magnesium Sulfate intramuscularly will cause convulsions.

EEG, CORTICOGRAPH, DEEP ELECTRODES

[redacted] makes extensive use of EEG and Corticograph equipment. In particular, brain wave studies are made in epilepsy and in shock therapy. They stated they were about to receive a new type of EEG machine which automatically registers in 16 positions. [redacted] separately stated that EEG and Corticographic studies would be very useful in testing new drugs in connection with their psychogenic effect. (Note: This statement has been made time and again to the writer. See report on [redacted] following.)

Along these lines, [redacted] and others commented that the new deeply inserted electrode techniques should be of immense value. Animal tests using these techniques are now being widely used and

r

results are most interesting and highly indicative. Dr. [redacted] felt that through these tests much new material on brain function would be developed.

ANIMAL EXPERIMENTATION

The writer discussed the use of animals and animal experimentation in testing and research on psychogenic [redacted] chemicals. [redacted] in the Medical Sciences Building and at a special area several miles outside of [redacted] operates extensive experimental animal colonies— these include many thousands of animals of many types including standard colonies of mice, rats, rabbits, pigs, dogs, cats, etc. but only a few (8 or 10) monkeys. These animals are primarily held for testing directly in connection with special clinical cases but pure research is continuously being conducted at all times with the animals. The writer was informed that the [redacted] would be glad to carry out animal experimentation using their animals if this were essential to some future work. (This offer was not developed pending future study.)

Dr. [redacted] stated that as far as he knew the best monkey colony in the U.S. was at the Institute of [redacted] under the directorship of [redacted]. [redacted] stated that Dr. [redacted] of [redacted] University was associated with [redacted] in this work and both were excellent men. He stated that the monkey colony at the Institute of [redacted] numbered always between 90 and 100 monkeys. He felt that both Dr. [redacted] and Dr. [redacted] would co-operate with us any time they were approached. [redacted] knew of no work at [redacted] with "waltzing mice" or with the Gibbon but expressed interest in studies with the Gibbon although admitting he knew little of the type.

From several conversations it appeared obvious that [redacted]

would be interested in the Gilbert but implementation of this idea was not developed.

NOTE: All personnel interviewed at [redacted] very emphatically stated to the writer that [redacted] and that experimental work could not be attempted with their patients except where it could be shown that it definitely worked for the patients good. They made it very clear that their work was to help those that were ill and pointed out that their specialties were all aimed in that direction. They said, however, specifically that this did not exempt them from studying the effects of the new drugs, determining possible uses of new drugs, designing instruments or from working to help the U.S. Government in any way they could. They stated their thoughts would have to be rearranged to meet the challenge of the various ideas the writer had expressed to them and that certain mechanisms could be set up along these lines and certain individuals should be cleared.

(see following)

DENTISTS AND DOCTORS RECOMMENDED FOR CLEARANCE

In the final discussion with Dr. [redacted], Dr. [redacted], Dr. [redacted] and Dr. [redacted] at [redacted] the following names were recommended for clearance by this Agency so that their particular skills and talents could be exploited. All of these individuals are described as loyal and patriotic American citizens and who, in the opinion of the doctor's present, would be willing to help and could make substantial contributions to various fields of endeavor which they felt the Agency

must be interested in.

Dr. [redacted]

[redacted] is a distinguished woman biophysicist and physicist. She has a vast range of interests; is regarded as exceptionally brilliant and is a specialist on the effects of ultrasonics, radar and radio on human beings.

Dr. [redacted]

[redacted] has been for a number of years a Research Consultant at [redacted] in biochemistry and pharmacology. Prior to that he was head of the department and Dean of the College of Pharmacology at the [redacted] University of [redacted] He is regarded as an outstanding pharmacologist.

Dr. [redacted]

[redacted] is Chief of the Department of Biophysics at [redacted] and regarded as one of the most brilliant physiologists in the United States today. [redacted] It is believed he probably has a clearance of some type IX from this Agency.

Dr. [redacted]

[redacted] is a consultant in Neurology to [redacted] and was recommended because of his knowledge of brain problems and related

matters and his recognized ability as a nerve specialist.

Dr. [redacted] is a consultant to [redacted] in physiology and medicine. More particularly, however, Dr. [redacted] is an experimentalist and researcher of considerable ability and has designed a great number of instruments which are believed to be of considerable interest to the Polygraph Branch. He has built many different types of Pleythsymographs and devices for the measurement of the flow of blood and blood pressure. Dr. [redacted] is described as a man of great imagination and a person who would be valuable for Agency use along lines of interest to us.

DR. [redacted]

Both of these men are consultants to [redacted] and are of considerable interest to us. Dr. [redacted] is an inventor and designer of special medical devices and has worked with Dr. [redacted] and others developing special anesthetic devices. Dr. [redacted] himself is the ranking doctor in anesthesiology at [redacted] is senior consultant.) According to Dr. [redacted], Dr. [redacted] is in the process of clearance now.

* * * *

As noted above, the visit to [redacted] was, in the opinion of the writer, very profitable, not so much for the fact that the writer was able to pick up new information or new chemicals but primarily because of the particular enthusiasm and willingness to co-operate shown by the individuals interviewed at [redacted] In proposing the names above for clearances, the ideas seems to be (this has the wholehearted approval

and support of [redacted] and these names plus the individuals contacted during this trip should be cleared at the Top Secret level for any type of contact or work on a priority basis. This group would then form a sort of pool of brain power or a nucleus that would work if required on problems of interest to this Agency. They would be a ready reserve fully cleared in the event of future trouble that would probably be extremely valuable in research and in advice.

The writer wishes to point out that this is not his idea but that it seemed to stem from the group. Certainly it could be said that the full facilities of the [redacted] can be made available for our work, with the only provision that it must be remembered [redacted] is a hospital and human experimentation except within very limited ranges is not carried out on private patients. No financial discussions were involved in any of the above and it was very emphatically pointed out to the writer that the [redacted] accepted no money during World War II for any of the work that was done for the U.S. Government, particularly for the U.S. Navy. As noted previously this trip was more of an exploratory nature and was not aimed at "immediate results" but was rather aimed at a possible full cooperation in the future.

SPECIAL COMMENTS

Of particular interest is the intercommunication system that is operating in [redacted] today. This is based upon a remarkable pneumatic tube system which covers the entire areas and all buildings and which operates with great speed and efficiency between the Headquarters Buildings and the other hospitals and even to [redacted] which is over one mile away. These tubes, according to [redacted]

are very secure and very fast. The security of the tubes is predicated on the fact if they are tampered with or opened, the tube ceases to function. In opinion, this development is extremely useful and very safe and saves a great number of personnel and much time. explained that it is essential that case histories, often bulky documents, must be sent rapidly between the various hospitals and must move immediately ahead of a visiting doctor. This is done through the development of a highly effective central communication system of pneumatic tubes, gravity pipes between floors and automatic lifts for horizonal movement. This central communications system set-up is the heart of the administrative operation of and related hospitals. In addition to the handling of documents, X-rays, etc. through the tube systems, it is also extremely interesting because of the general intercommunication techniques and the simple instrumentation developed for rapidly locating individuals and for the handling of the tremendous number of patients that flow through the hospitals. who is the civilian director of the stated to the writer, after the writer had expressed admiration and interest in the intercom system, that he would very gladly permit inspection and examination of the system by officials of this Agency and would cooperate in any way with the government if a study of the system would help. This same cooperative attitude was expressed by (Director, Medical Sciences Building) who informed that they would be welcome again and questions concerning the Medical Sciences Building having to do with the administration structure or working facilities of it would be more fully and completely explained at any time they so wished.

RE: [redacted] -- [redacted] -- LSD

By an extremely unusual co-incidence, [redacted] of [redacted] (the manufacturers of LSD) and the [redacted] representative were visiting the [redacted] for business reasons during the two-day period the writer was present. Of particular interest was the fact that [redacted] had appointments with both [redacted]. Because of this, and because of the extremely favorable attitude of [redacted] and [redacted], the writer requested that [redacted] be questioned concerning LSD and certain other related matters. These conversations, reduced to the simplest terms, reflect that [redacted] would give to [redacted] any amount of LSD required for experimental research and that [redacted] was still extremely interested in the material and was still producing same. As to the interests of the USSR in LSD, it came out in a conversation between [redacted] and one of [redacted] representatives that [redacted] felt that the USSR was not working with LSD and apparently were not at all interested in the drug. [redacted] stated he based this on the fact that several years ago he attended an international conference, apparently in Paris, and at this conference there were present a number of Soviet scientists and somehow or other the subject of ergot and its derivatives was brought up. The Soviet scientists explained to [redacted] that they wanted no part of ergot; they had spent a good deal of time and effort stamping out the rusts that affected the grain in the Soviet Union (ergot derives from a rust) and they had no interest in the chemical (LSD). [redacted] Apparently felt and was satisfied that this was indicative of their attitude and interest. [redacted] also informed [redacted] that he was going to the West Coast in the next few days to attend a "West Coast Federation

meeting at San Francisco. [redacted] also stated he is going to be at Atlantic City, New Jersey, to attend the American Psychiatric Convention from 9-13 May 1955. It should also be noted that [redacted] intends to attend this convention and further, for matter of record, [redacted] is to attend the convention and is chairman of one of the groups.

Of particular interest are the following statements which arose from conversations between the writer and [redacted] answers were based on his own views and his conversations with [redacted] stated that [redacted] was interested in the use of Chlorpromazine as an antagonist to LSD. Further [redacted] had not tried combination effects with LSD but they do intend to try. In addition, [redacted] is working on Seratonin. Seratonin is described as apparently an antagonist to LSD of the peripheral nerves.

Of particular interest to the writer was that [redacted] made the statement that LSD in a human body after administration is concentrated in the liver to the extent of 99% and less than 1% is found in the brain and this disappears extremely rapidly.

Also of extreme important was the fact that [redacted] made the emphatic statement that LSD is stable and can be used in aerosols, in ducks and can be administered through the skin. For matter of record, the writer brought this up several times to make certain he had heard [redacted] statements correctly. The writer, however, being non-technical, did not develop this nor press [redacted] for explanations nor did he request [redacted] to develop this theme with Dr. [redacted] It should be recalled that [redacted] as well as others contacted assured the writer they will attempt

to answer questions put to them in the future and [redacted] had indicated his willingness to convey any questions of this type to those individuals for specific answers.

While [redacted] was talking with [redacted] stated a prominent figure in Hollywood whose name was not mentioned (and [redacted] stated he did not ask) had written to [redacted] and asked for a quantity of LSD stating he wanted to use it at a party. [redacted] stated [redacted] was shocked at this and turned it down but subsequently learned this same person in Hollywood had obtained LSD and also Mescaline (sources unknown) and had used both in some sort of punch at a party and the psychotic effects and results were violent and terrible.

NOTE: The writer felt that he could not press this any further nor ask Dr. [redacted] to go back and attempt to obtain further information from [redacted] at this time. [redacted] stated he was curious as to the effects and who had the party and would try to pump this information out of [redacted] at Atlantic City in May.

NOTE: It should be recalled that [redacted] recently wrote a book on the effects of Mescaline and is knowledgeable about LSD and as far as the writer knows is living in Hollywood, California.

NOTE: The writer considered the idea of posing as a doctor at [redacted] and interviewing [redacted] with [redacted] but abandoned the idea in view of the fact that [redacted] might possibly become suspicious if too sharp questions were asked concerning LSD. [redacted] was described by both Mrs. [redacted] and [redacted] as a perfectly sincere, honest, and intelligent scientist.

In addition to the above, the fire control system of the new buildings is worthy of study. As in most modern industrial plants and in large modern office buildings, fire control and fire reporting signals are important. These buildings have a very elaborate centrally controlled fire spotting and fire signalling apparatus which is based on a central control panel in a special area. Duplicate panels are placed in strategic points elsewhere in the complex to aid fire apparatus and aid anyone reporting or fighting fires. This central panel also contains a great number of indicators, dials, regulators for temperature, air pressure, air conditioning, etc. While not unique, this would nevertheless be worthy of study of any person visiting this area for the special reasons set out above.

Also of additional interest was a 180-seat theater that was specially designed for lectures, movies, demonstrations or whatever purpose required. It apparently is accoustically perfect and by special arrangment of the seats the stage (or things on the stage) are clearly visible from the seat area. It, of course, is equipped with the latest movie and sound devices and an excellent PA system. It is obvious that this Agency, if it moves into new quarters, must have theaters of this type built in and it is believed that a survey of this theater might be valuable for our purposes.

Another item which obviously should be of interest to Security and to other elements of the Agency was the fact that practically all of the working areas of the new buildings are equipped with many different types of facilities built in. For instance, in the research areas, there are plugs in the wall for automatic timing and there are sockets for light and heavy power. There are oxygen compressed air, high and low

pressure steam, hot and cold water taps and other gas outlets and there were vacuum tubes in many places as simple wall outlets. This type of wiring and piping in the opinion of the writer is also worthy of survey since in highly sensitive areas in any new building which this Agency must have some of the these elements or methods of installation might be very important.

Paragraph No. 2

On Saturday, 9 April, the writer talked to Dr. [redacted] at [redacted] Hospital, between 2:00 and 5:00 PM. [redacted], as usual, was most co-operative and again expressed his willingness to carry out any type of work or testing that would be possible under his present circumstances. He stated that his work precluded exhaustive research but felt that these weapons ("truth serum" and psychogenic chemicals as intelligence weapons) were ones that required a great deal more time and attention than anyone he knew of had placed in them to date. He stated at the present time his criminal work with [redacted] (Director of Security at the [redacted]) was slacking off because [redacted] was not in good health and because certain people at the [redacted] regarded the use of the "truth serums" techniques, even for criminal cases, unsuitable as a [redacted] function. (Other sources indicate [redacted] is in bad repute at the [redacted] and may be discharged.)

[redacted] said he had done a few more cases and his results had been good as far as the police were concerned. [redacted] stated at the present time he either used scopalomine intravenously at the beginning of the technique or seconal. Thereafter, he uses pentothal immediately or at

least within the next ten minutes. He said he considered this safe and it has produced satisfactory results as far as he was concerned.

Dr. [redacted] stated he had not had a chance to test out Chlorpromazine with pentothal and felt this should be done. He also suggested that we examine a new Eli Lilly chemical Valamin or Valarid.

Dr. [redacted] stated he had interest in this work, there were an infinite number of chemicals to be tried and in his opinion (stated to the writer previously) an EEG machine was essential to this operation. [redacted] stated in in the near future, he was to have an EEG machine for the anesthesiology department at the hospital for their sole use and with this he could and would test in a limited way if we would send chemicals of interest to us with proper supporting information.

Dr. [redacted] stated that we should examine a new chemical called NP13 which has been developed by an Australian in Australia by the name of Dr. [redacted] (fnu). Dr. [redacted] described this chemical as a specific antagonist to the barbiturates and he has been attempting to obtain the chemical from Dr. [redacted]. He stated [redacted] had promised to send some of the chemical to [redacted] for experimental purposes and [redacted] stated he would work with it when it arrived and keep us informed. In this connection he stated there was an article appearing in the LANCET of January 1955 and the BRITISH JOURNAL, January 1955, concerning this new chemical.

Dr. [redacted] stated that some time ago he had attended a convention in the East, apparently New York City, at which [redacted] was present. [redacted] stated he knew of [redacted] work in anesthesiology and that he has, of course, suspected that [redacted] was engaging in work

[redacted]

along "truth serum" lines. He stated that [redacted] invited him up to his room and that a discussion was held during which [redacted] asked [redacted] how his work was progressing on "truth serums." According to [redacted] asked, "How did you know about that?" and apparently denied working on it. [redacted] told [redacted] "I can read between the lines" and when he pressed the subject, [redacted] apparently told [redacted] he had another conference to attend and immediately left the room. In this connection, it is interesting to note that [redacted] stated that [redacted] was generally regarded as a "pansy" by [redacted] who knew him but [redacted] said [redacted] did not make a pass at him, although [redacted] remarked that [redacted] asked him if he would like to take a shower while he was in the room.

No other details along this line were developed.

[redacted] suggested that the following names be placed in the Agency confidential files: He stated one [redacted], presumably a doctor, did not have his contract renewed at the University of [redacted] because he is apparently a Communist. He stated that [redacted] when he knew at the University of [redacted] and also a doctor, had refused to go [redacted] to the Veterans Administration to accept a position because he had to sign an oath.

[redacted] suggested that [redacted] was a very high type of person and should be cleared for contact. [redacted] thought [redacted] had some sort of government clearance anyway.

For the record, it should be noted that most of the discussion concerned matters previously taken up with [redacted] and which have been reported previously. As stated, [redacted] was extremely co-operative and offered his services

[redacted]

in any way he could be of help to us. In this connection, [redacted] pointed out that his wife had died suddenly last fall and since he has no children he had not yet settled on his immediate future having several offers for work elsewhere which he was considering. He repeated that if we would send him any chemicals with the necessary information for their use he would be glad to make limited tests with them along lines of interest to us. He requested we forward information on Serotonin to him and also send him what material was available on the work that [redacted] was doing at [redacted] was acquainted with some of the literature on LSD but told the writer he had not worked with it experimentally or in any of his clinical cases. He recognized its extreme interest from a psychogenic point of view and suggested it required experimentation, particularly with other chemicals.

NOTE: In the writer's opinion, we should forward to [redacted] certain chemicals for his study and use but we should not go much beyond this at the present time. The writer feels that [redacted] loss of his wife may, at the present time, still be too fresh in his mind for him to become involved in work of a highly sensitive nature. While it is obvious that [redacted] was ready and willing to co-operate with us and in a small way we should send him some material, the writer feels that extensive commitments should not be made to him or sensitive material should not be sent to him now.

Visit No. 3

The writer spent Easter Sunday afternoon, 10 April and part of that

evening talking with [redacted] discussed the proposed work at [redacted] and stated as soon as clearances come through the entire operation would begin. He seemed very enthusiastic about it and informed the writer his team was interested and ready to go. He stated that if they were not overwhelmed by red tape or misdirected, they should produce extremely valuable information from either a negative or positive point of view. He stated [redacted] was an extremely brilliant man and was capable of producing ideas that should be of great value. He urged that the writer on his next trip through [redacted] arrange a conference with [redacted] employees. [redacted] was constantly working with and experimenting with new and unusual chemicals and combinations and the writer should talk to [redacted] and brief him on new chemicals and new areas of interest since [redacted] ideas would be valuable.

[redacted] is still very strongly oriented in our direction and his enthusiasm and energy are worthy of comment. As is well known, his police connections and other connections in the [redacted] are very wide and important and it is felt that [redacted] irrespective of the [redacted] operation for [redacted] should be constantly held as an SO consultant in case of XXX future need.

The writer informed [redacted] when he left him that he would make arrangements on his next trip to the area to sit down with [redacted] and [redacted] to discuss in detail the various angles of interest to the ARTICHOKE work and its relation to the [redacted] Project.

On Monday, 11 April, between 2:00 and approximately 5:00 PM in the [redacted]

afternoon, the writer talked to [redacted] Professor of Mycology at the University of [redacted] who is regarded by many knowledgeable individuals as probably the outstanding mycologist in the United States, if not the world, today. He was extremely co-operative and surprised the writer by asking after the general subject had been introduced, why the very fertile field fo toxic mushrooms had not been further explored. He said he was well aware of the fact that there grew in the Soviet Union mushrooms that were toxic and the literature contained many examples of the Russians, particularly the primitive Siberian tribes, using mushrooms as intoxicants.

The writer showed the researcher the very extensive list of questions and names of various mushrooms which Dr. [redacted] had sent with the writer. He examined the list and stated that the list was accurate and interesting but stated there were literally thousands of mushrooms in the United States about of which we know nothing and hundred of which are toxic. He stated if work were to be done along these lines, these species that exist in the U.S, and Canada should be studied first. He stated that the toxicity or chemical content of the mushroom is not in his field and this would require a pharmacologist or chemist plus careful research. He urged, however, that this work be carried on.

He stated that this summer he was going to take a field trip collecting specimens [redacted] and stated he would be very glad to collect and preserve specimens for us. He stated he would do this free but if extensive collections were to be carried out, a small sum of money would be needed to hire assistants to help him collect (probably several hundred dollars would be sufficient). He said he would make the

po

positive identifications and direct the research for us. [redacted] then stated in 1956 he was taking a fifteen months field trip working in western U.S. (California and Puget Sound areas) on a $12,000 grant from the [redacted] and stated he couldn't see why we couldn't be included in a grant of this kind and why he couldn't collect material of interest to us on this field trip. In developing his ideas of work on ways he could assist this Agency he felt before anything else should be done and certainly before he leaves for summer work this year, someone from the Agency entirely familiar with mushrooms and who understands the problems involved technically should visit him. He stated he would be glad to sit down and discuss the entire problem with that individual and would welcome ideas and thoughts along lines of interest to us.

[redacted] then stated emphatically to the writer that he regarded the following as a matter of extreme intelligence importance. He stated that from his study of mushrooms and from his knowledge of Soviet activities along these lines he is completely convinced the Soviets are attempting to produce latex or rubber from mushrooms. He stated specifically that the Bulgaria Rafa has a rubbery fruiting body and he believed that the USSR was interested in this mushroom for rubber. He stated in addition to this that the Lactaria group of mushrooms definitely had latex in its body and that he knew that the Soviets would be interested in this group. He pointed to the fact that if the Soviet Union was able to develop strains of these mushrooms that could be grown in vats or indoors or under controlled circumstances and could come up with a way of developing this successfully, they would have a source of crude rubber and it was his guess they are doing exactly that. He stated that this was extremely important. He suggested th

that Dr. [redacted] of the Botanical Institute of the University at [redacted] was an outstanding mycologist and would know quite well of U.S.S.R. activities in this field. He stated he did not know the political orientation of Dr. [redacted] but he should be contacted if possible.

[redacted] called attention to [redacted] address in [redacted] Denmark. [redacted] said [redacted] had worked with him at one time and was a capable mycologist but undoubtedly a Communist.

In addition to this, [redacted] said he was now convinced [redacted] government was studying and collecting mushrooms, not for agriculture reasons, but for other reasons. He stated up until a few years ago there was no interest in the Finnish government in mushrooms even though they grow there abundantly. He stated now there is a branch of the [redacted] Department of Agriculture [redacted] that is very active in this field and stated this department is headed by Dr. [redacted] who is a capable mycologist and taxonomist. He stated it was of particular interest to him because a young lady had been sent by Dr. [redacted] from the Department of Agriculture [redacted] to work during the preceding summer with [redacted] This young lady, whom [redacted] regarded as very capable, was asked by [redacted] why she was working with him in [redacted] and not looking for mushrooms in northern [redacted] The young lady replied that is not my branch, others are covering those areas. This indicated to [redacted] that extensive work was being done along these lines in [redacted]

Dr. [redacted] urged that we clear [redacted] of the Department of Botany of the University of [redacted] He stated [redacted] is a reliable American citizen, a very capable mycologist and highly recommends him. He called attention to the fact that the Puget Sound area contains a great many toxic

mushrooms including many Amanita Pantherina.

Dr. urged that [redacted] of the Biology Department of [redacted] College be cleared and highly recommended him as an excellent mycologist. Dr. pointed out that there were many well-known and many unknown toxic mushrooms existing between Florida and Texas and [redacted] was acquainted with this territory intimately.

Dr. requested that we check out Dr. who is an officer of the [redacted] He stated this organization, of which he knew nothing, is of extreme interest to him because Dr. had approached him in connection with a rubber-producing mushroom.

Dr. requested that Dr. Director of the University of [redacted] who is technically [redacted] boss, be cleared. He stated that if he were doing work for us either officially or unofficially, [redacted] should be advised of it although not necessarily in detail. Dr. stated as far as he knew, Dr. had a government clearance of some type.

Dr. stated he knew Dr. who is now working at [redacted] Argentina. [redacted] stated [redacted] is a capable mycologist but his political background is dubious, probably Communistic.

Also of interest is the fact that Dr. a mycologist [redacted] had recently visited him and asked him concerning mushrooms producing psychogenic effects. Dr. said he was quite polite to [redacted] but had to tell him as he told the writer that the field was enormous and required a much more specific approach before he could help. Dr. said he did not know [redacted] but from the nature of the conversation, he felt that [redacted] was beginning a study of mushrooms having psychogenic effects.

[redacted] stated he had been eating mushrooms all his life and he had had mushrooms make him sick, make him dizzy, destroy his coordination, and orientation and, in fact, act upon him in almost every way except kill him. He stated he had tested mushrooms of all varieties and run across numerous mushrooms that had many mental and physical effects. He gave examples of people having eaten mushrooms and falling from bicycles and he and his wife after eating Verpa Bohemia, a small wild spring variety, apparently native to [redacted] had had their visual coordination completely upset and he himself after eating this mushroom had walked into a wall while trying to walk through a door. Because of these, [redacted] stated this whole field should be explored. He again reiterated that he himself was not in a position to make chemical tests or pharmacological tests on mushrooms but he could collect and identify mushrooms along lines which were of interest to us and he would do so if properly requested and the necessary technical arrangements were made with him.

For the record, it should be noted that [redacted] listed numerous mushrooms of interest for the writer and commented on others. [redacted] also showed the writer the collection of fungi and related plants at the [redacted] and in addition showed the writer the correspondence files on mushrooms, which covers a worldwide net.

In the writer's opinion, [redacted] appears to be extremely valuable and a future source of information for the Agency and should definitely be exploited. The writer recommends that someone having technical knowledge visit [redacted] as soon as possible and that the question of financing [redacted] should be taken up probably with [redacted] at once. While the writer has no knowledge of the Agency's connections with the [redacted] he is certain such connections exist and the writer recommends that full

advantage be taken of the fifteen month collecting trip that Dr. Wills will undertake in 1956 for the [redacted] and the University of [redacted]